Chapter 4

IP Addresses: Classful Addressing

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The address space of IPv4 is 2^{32} or 4,294,967,296.

Binary notation:

01110101 10010101 00011101 11101010

Dotted-decimal notation:

01110101 10010101 00011101 11101010

117 149 29 234

117.149.29.234

Hexadecimal Notation:

75 95 1D EA

0x75951DEA

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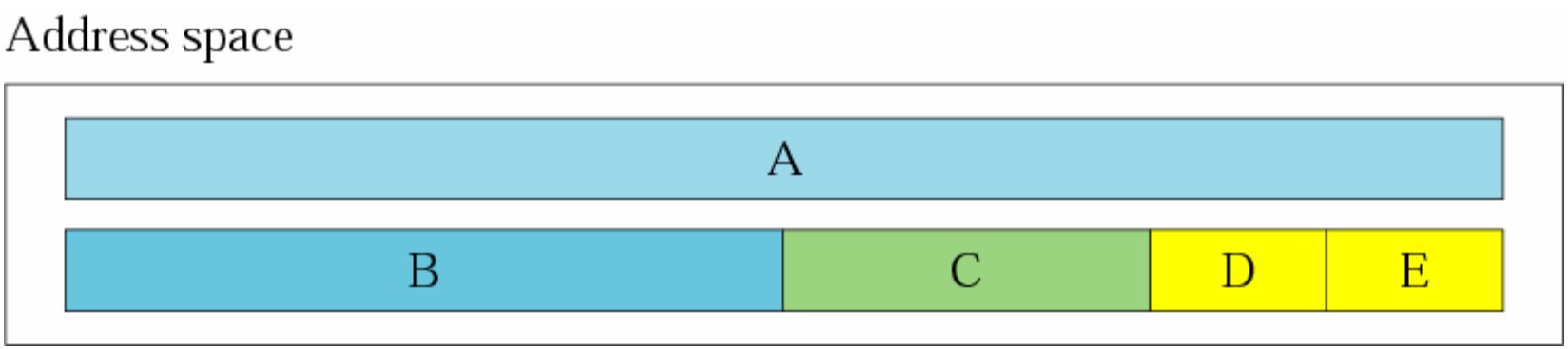
Occupation of the address space

In classful addressing, the address space is divided into five classes: A, B, C, D, and E.



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Finding the class in binary notation

	First byte	Second byte	Third byte	Fourth byte
Class A	**0**			
Class B	**10**			
Class C	**110**			
Class D	**1110**			
Class E	**1111**			

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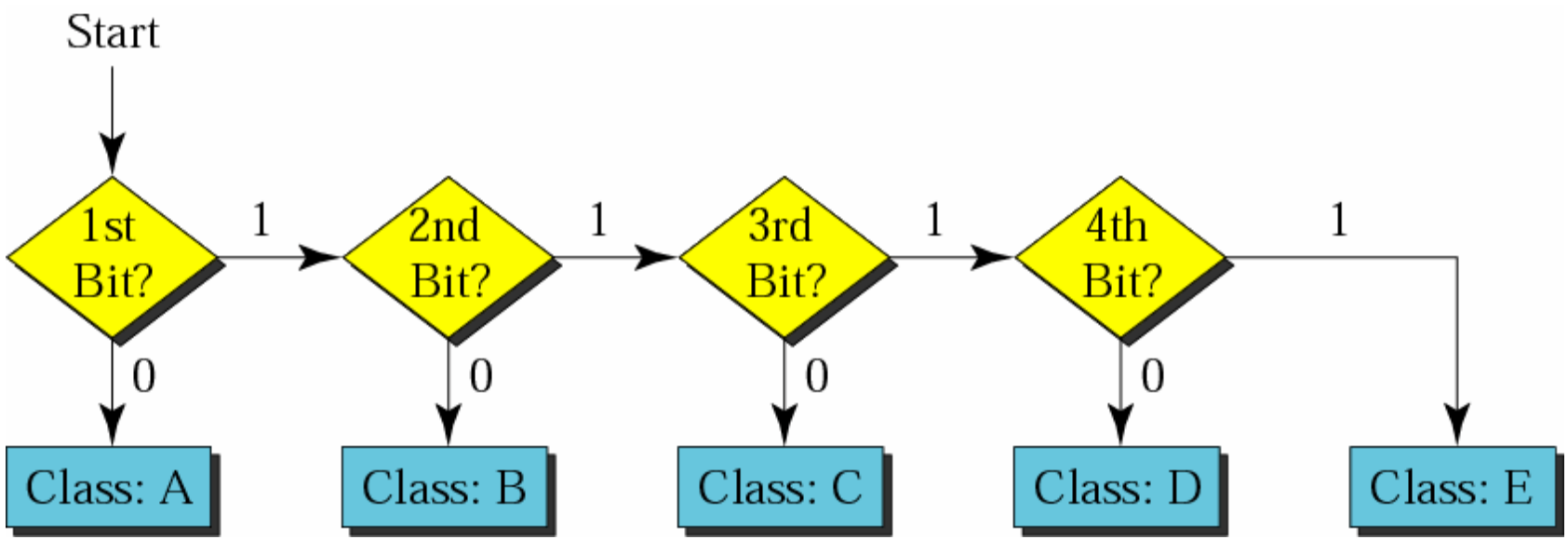
Finding the address class



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Finding the class in decimal notation

	First byte	Second byte	Third byte	Fourth byte
Class A	**0 to 127**			
Class B	**128 to 191**			
Class C	**192 to 223**			
Class D	**224 to 239**			
Class E	**240 to 255**			

For example, class B has min address (first byte) $10000000_B = 128_D$

max address (first byte) $10111111_B = 191_D$

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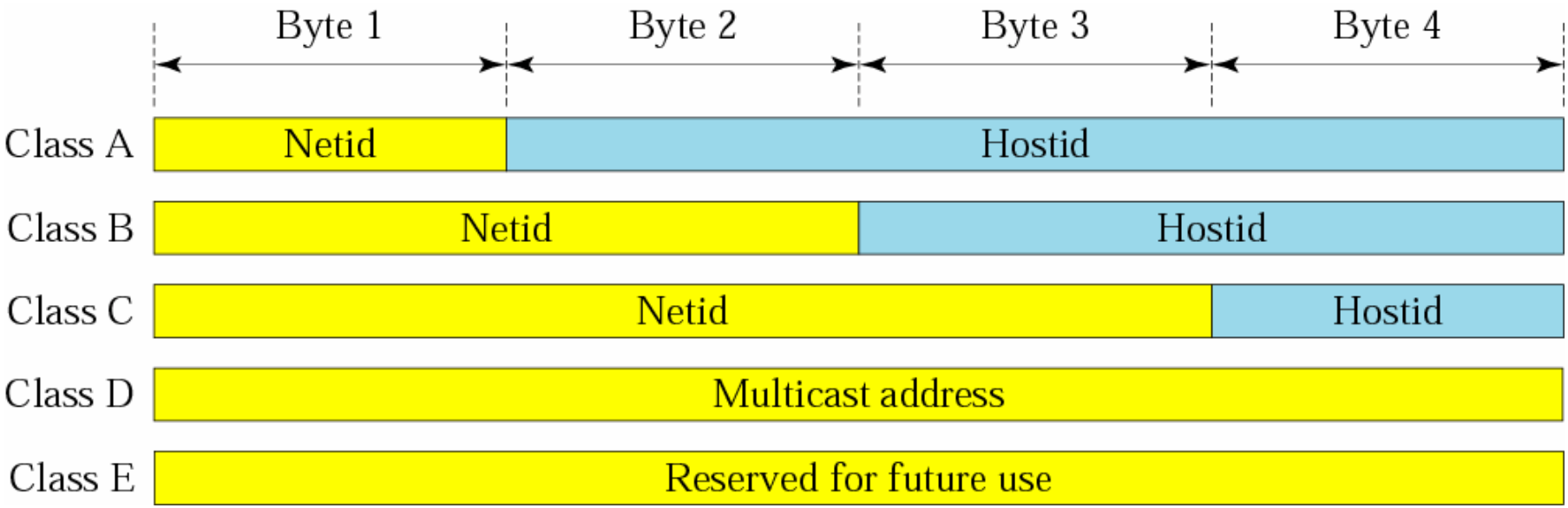
Netid and hostid



For example, class B has $2^{8-2} \times 2^{8} = 16{,}384$ networks,
Each network has $256 \times 256 - 2 = 65{,}534$ hosts
(addresses with host ID = all zeroes, or all 1s are not used for hosts)

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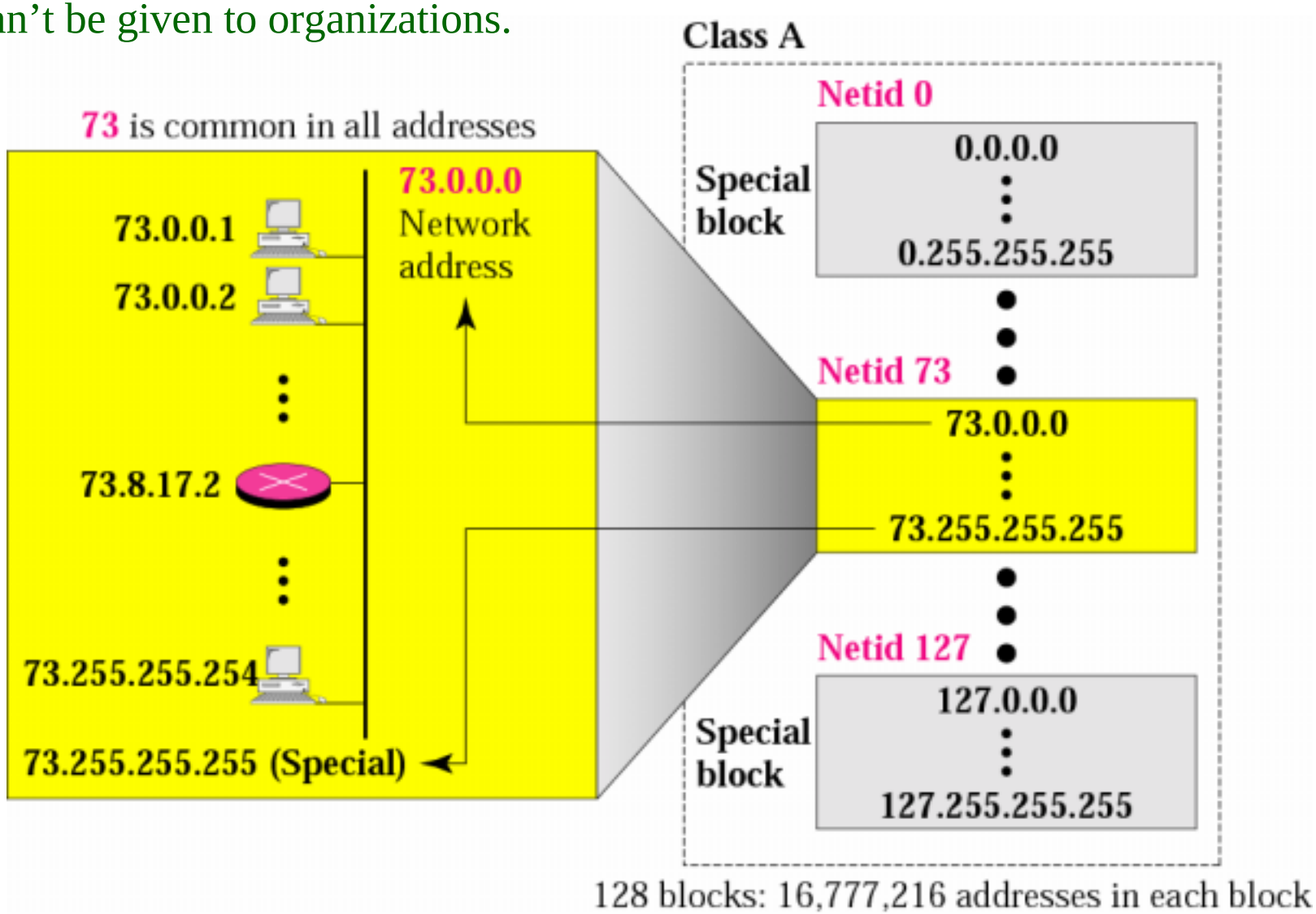
Blocks in class A

Net addresses 0, 10 and 127 are special and can't be given to organizations.



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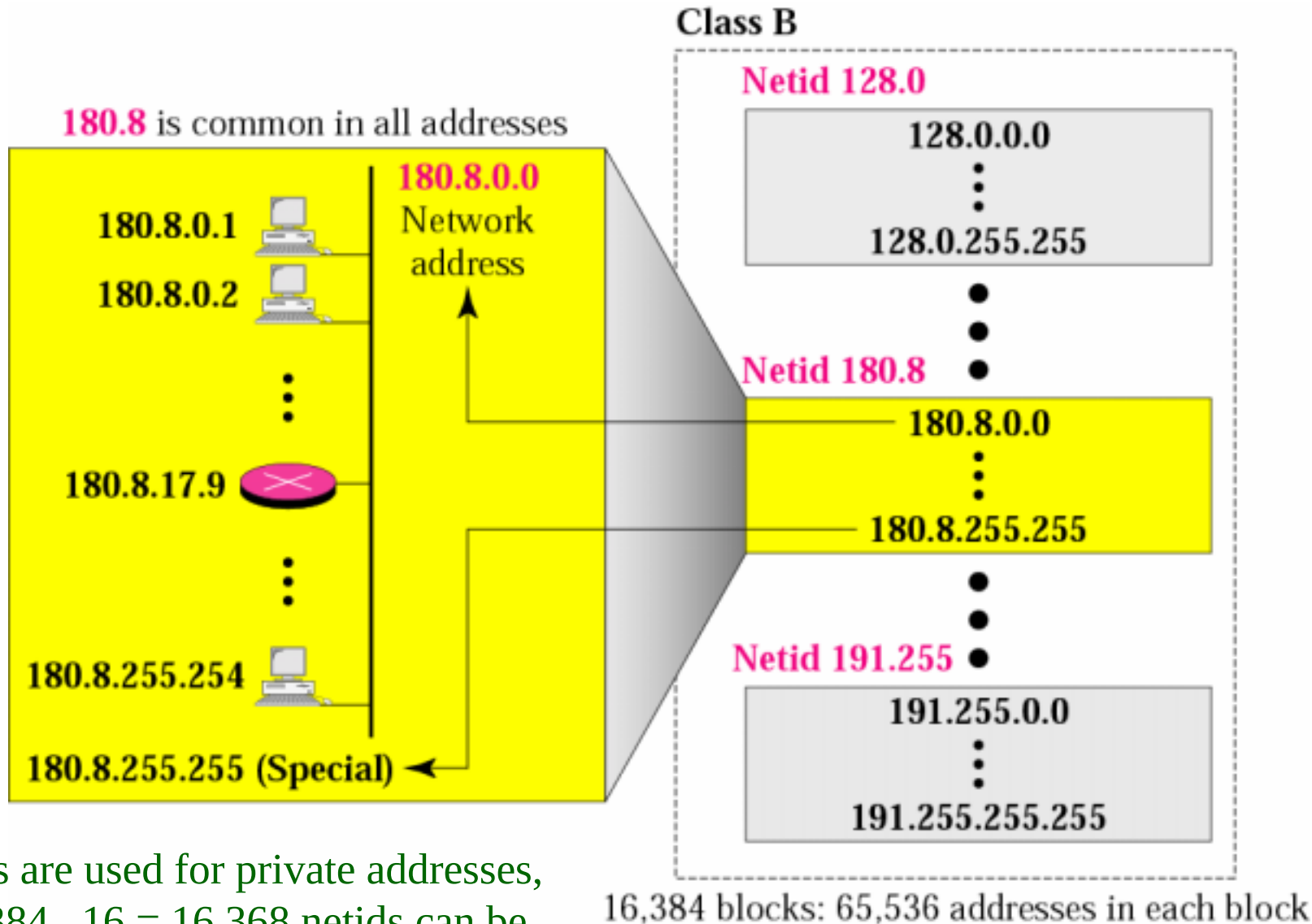
Blocks in class B



16 blocks are used for private addresses, only 16,384 –16 = 16,368 netids can be assigned to organizations

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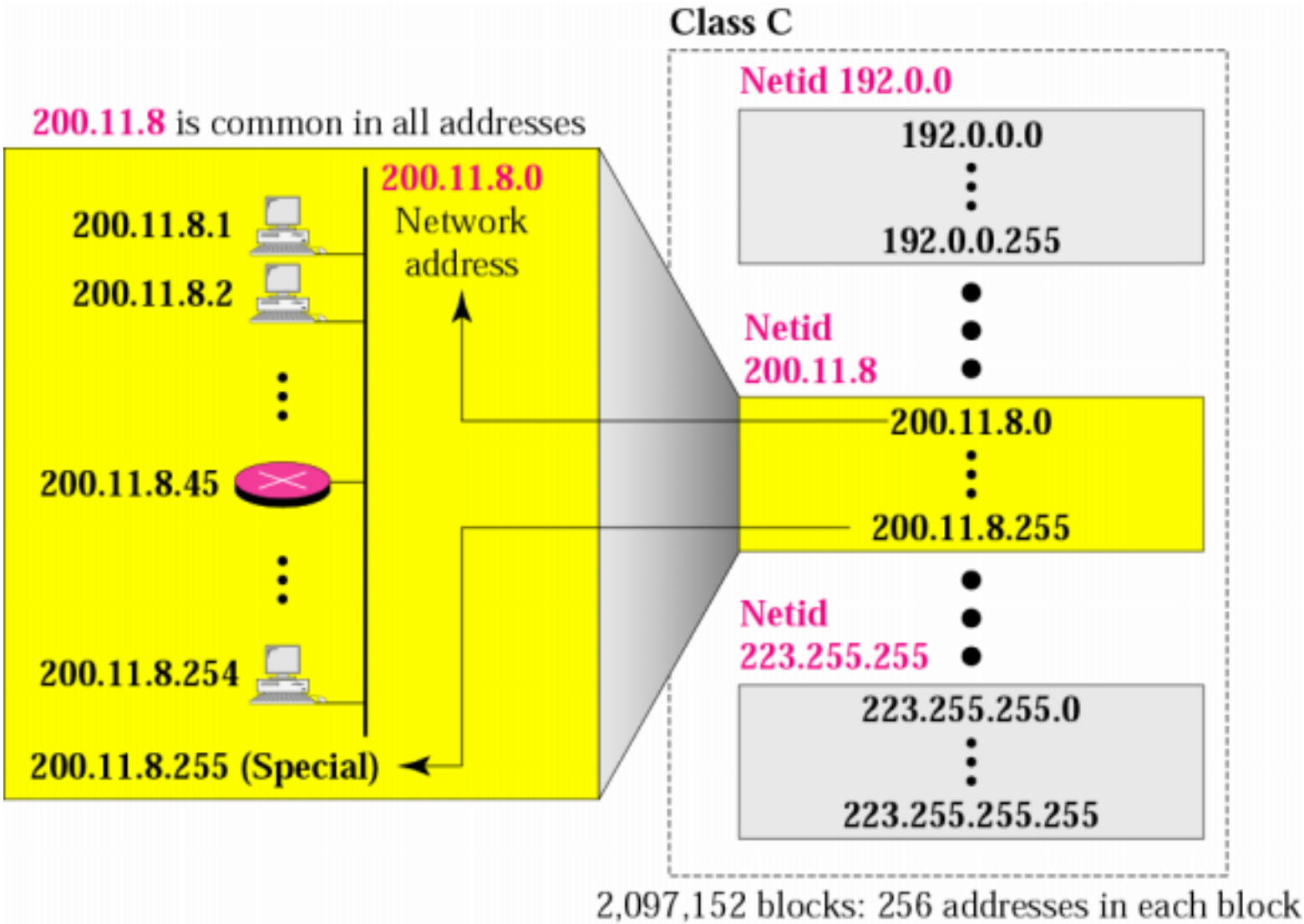
Blocks in class C



Class C has (223-192+1) x 256 x 256 = 2,097,152 netids

Each block can have 256-2 = 254 organizations

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Network Addresses

The network address is the first address.

For example the network address of 130.191.27.100 is 130.191.0.0

The network address defines the network to the rest of the Internet.

Given the network address, we can find the class of the address, the block, and the range of the addresses in the block

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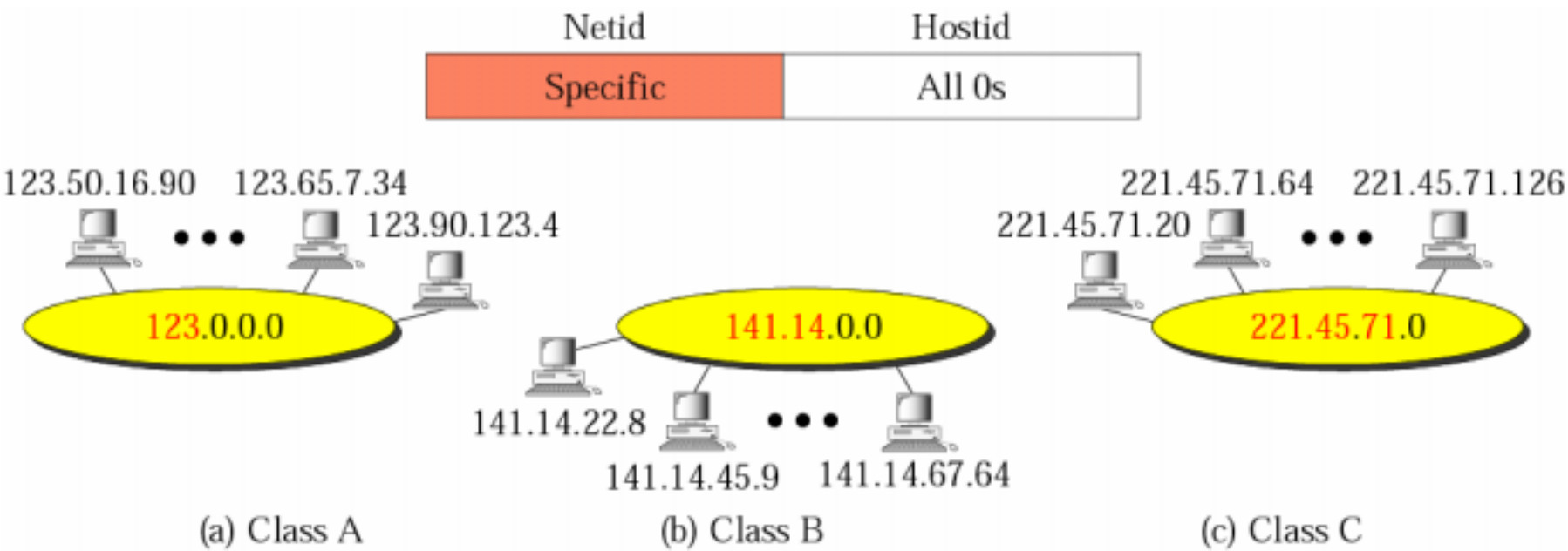
Examples of network addresses



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Example 9

Given the network address 17.0.0.0, find the class, the block, and the range of the addresses.

Solution

The class is A because the first byte is between 0 and 127. The block has a netid of 17. The addresses range from 17.0.0.0 to 17.255.255.255.

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Example 10

Given the network address 132.21.0.0, find the class, the block, and the range of the addresses.

Solution

The class is B because the first byte is between 128 and 191. The block has a netid of 132.21. The addresses range from 132.21.0.0 to 132.21.255.255.

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Example 11

Given the network address 220.34.76.0, find the class, the block, and the range of the addresses.

Solution

The class is C because the first byte is between 192 and 223. The block has a netid of 220.34.76. The addresses range from 220.34.76.0 to 220.34.76.255.

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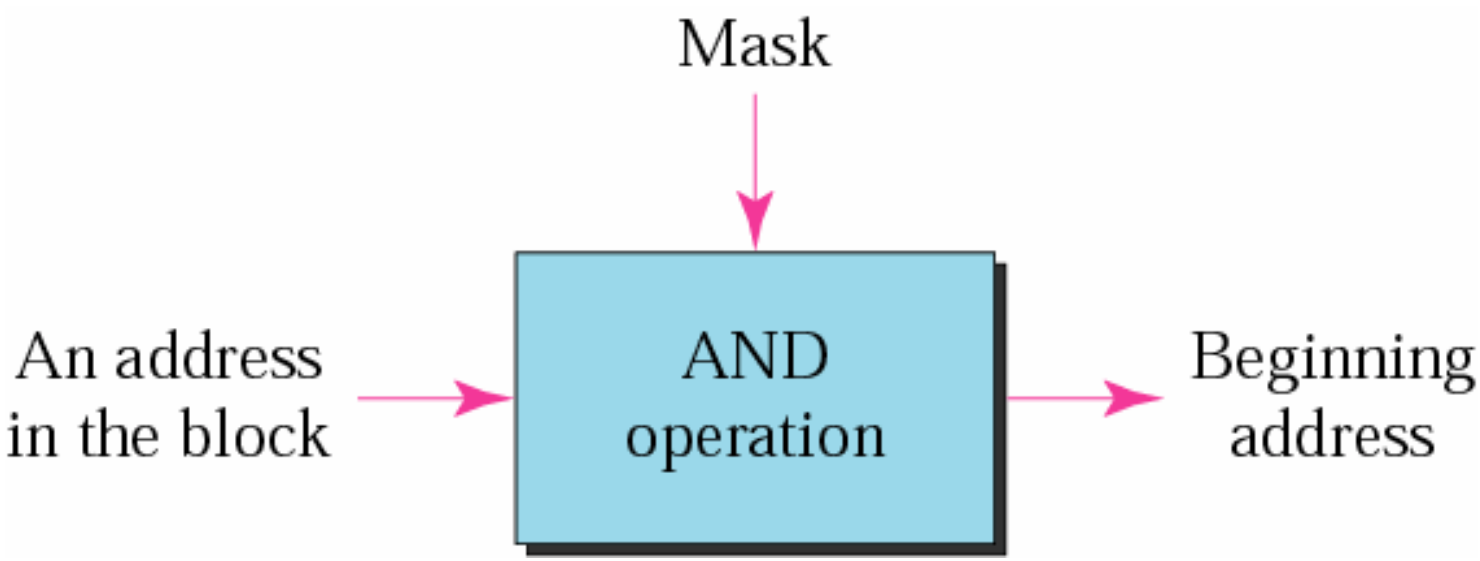
Mask

A mask is a 32-bit binary number that gives the first address in the block (the network address) when bitwise ANDed with an address in the block.



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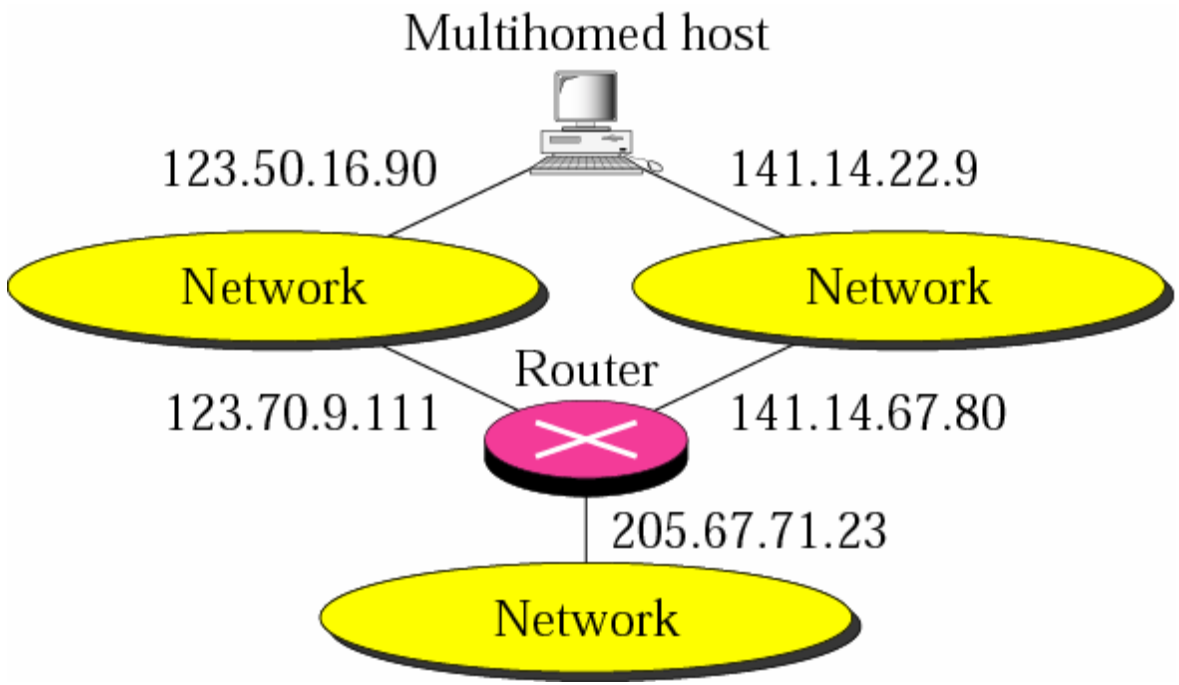
Multihomed devices



Multihomed hosts or routers have more than one interface and more than one IP address (which are not necessarily of the same class)

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Private Addresses

A number of blocks in each class are assigned for private use. They are not recognized globally.

Class	*Netids*	*Blocks*
A	10.0.0	1
B	172.16 to 172.31	16
C	192.168.0 to 192.168.255	256

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Special Addresses

Type	*Netid*	*Hostid*	*Source/Dest*
Network address	xxxxx	00000	
Direct b/c address	xxxxx	11111	Destination
Limited b/c address	11111	11111	Destination
This host on this net	00000	00000	Source
Specific host on this net	00000	xxxxx	Destination
Loopback address	127	xxxxx	Destination

(b/c – broadcast)

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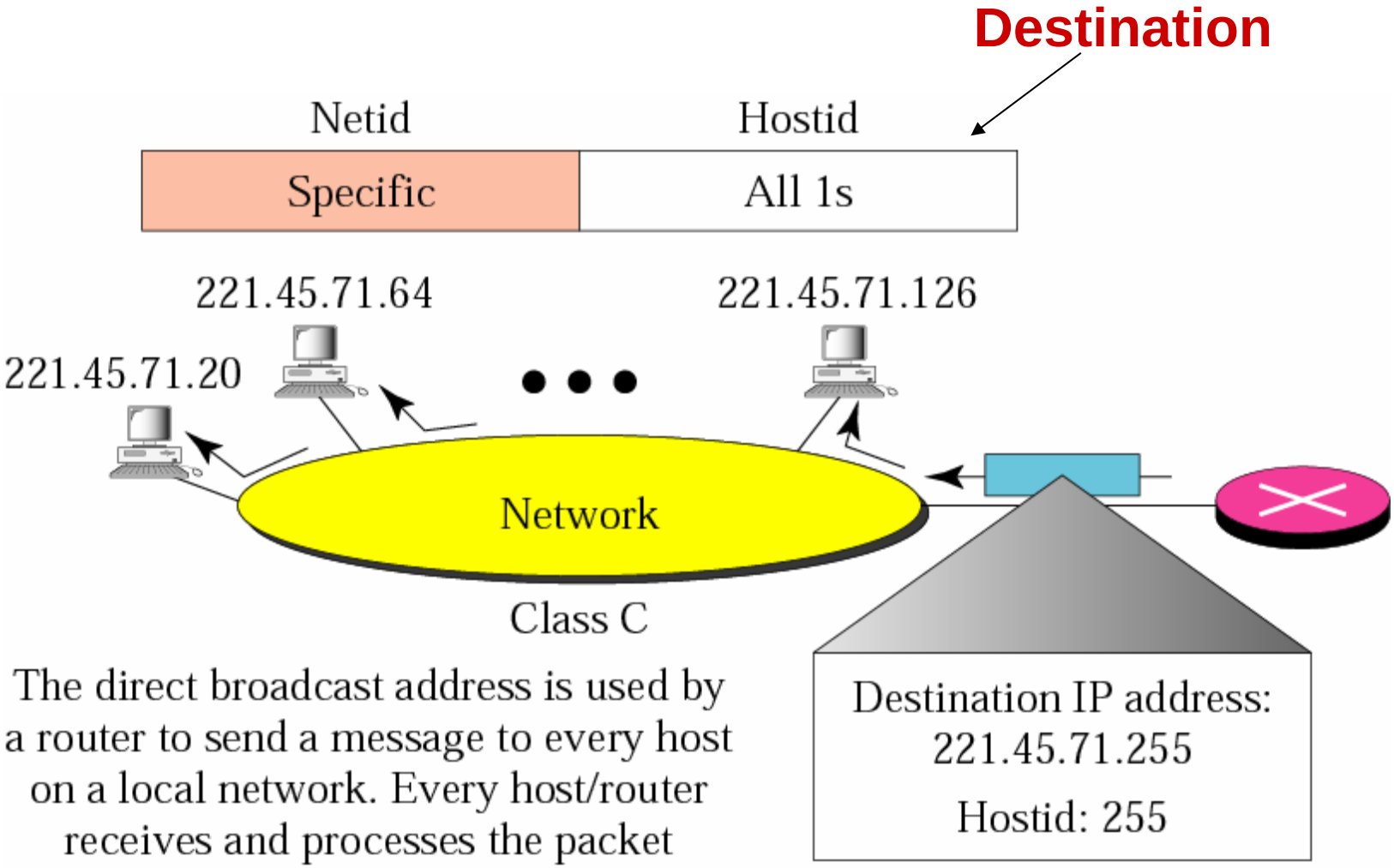
Example of direct broadcast address



The direct broadcast address is used by a router to send a message to every host on a local network. Every host/router receives and processes the packet with a direct broadcast address.

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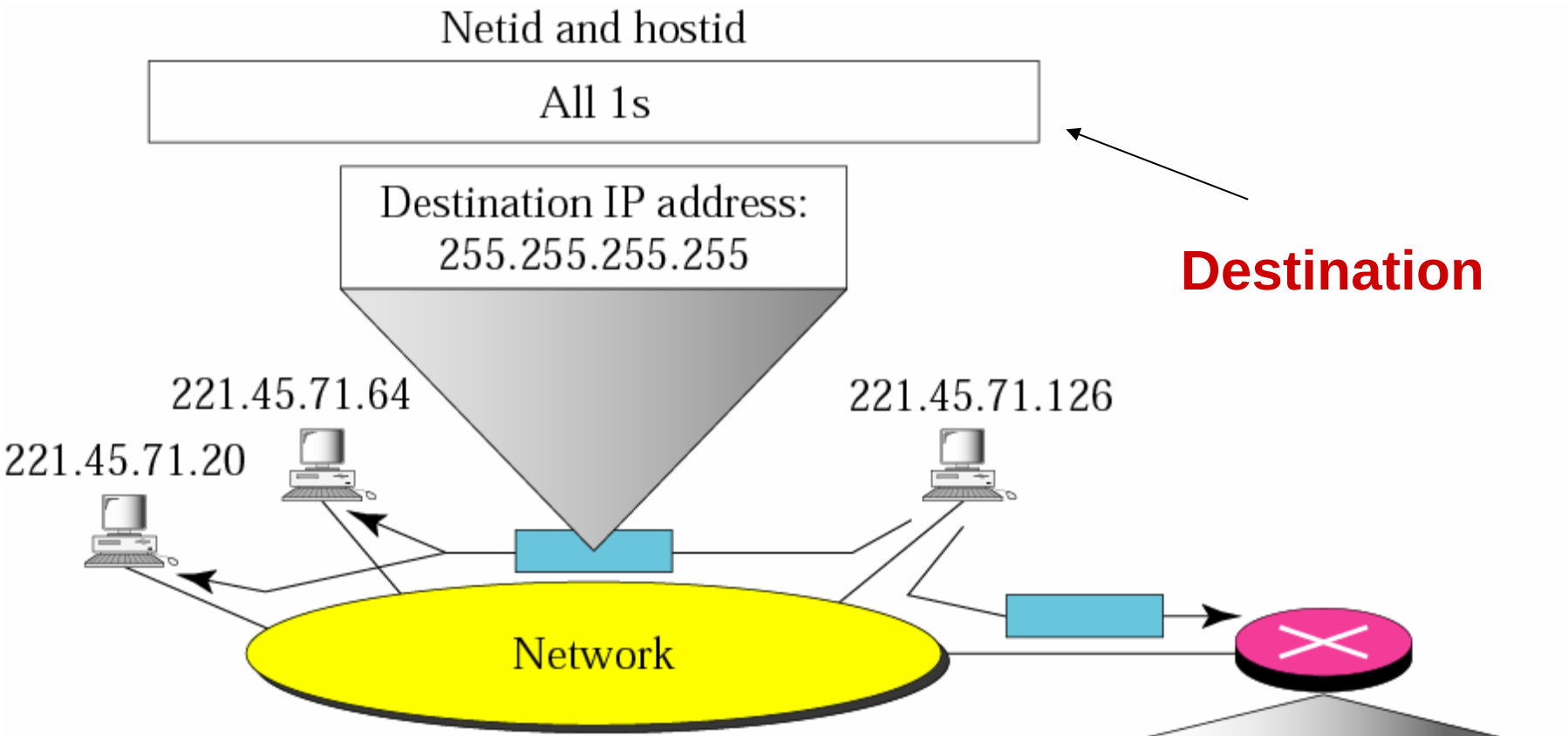
Example of limited broadcast address



A limited broadcast address is used by a host to send a packet to every host on the same network. However, the packet is blocked by routers to confine the packet to the local network.

Router blocks the limited broadcast packet

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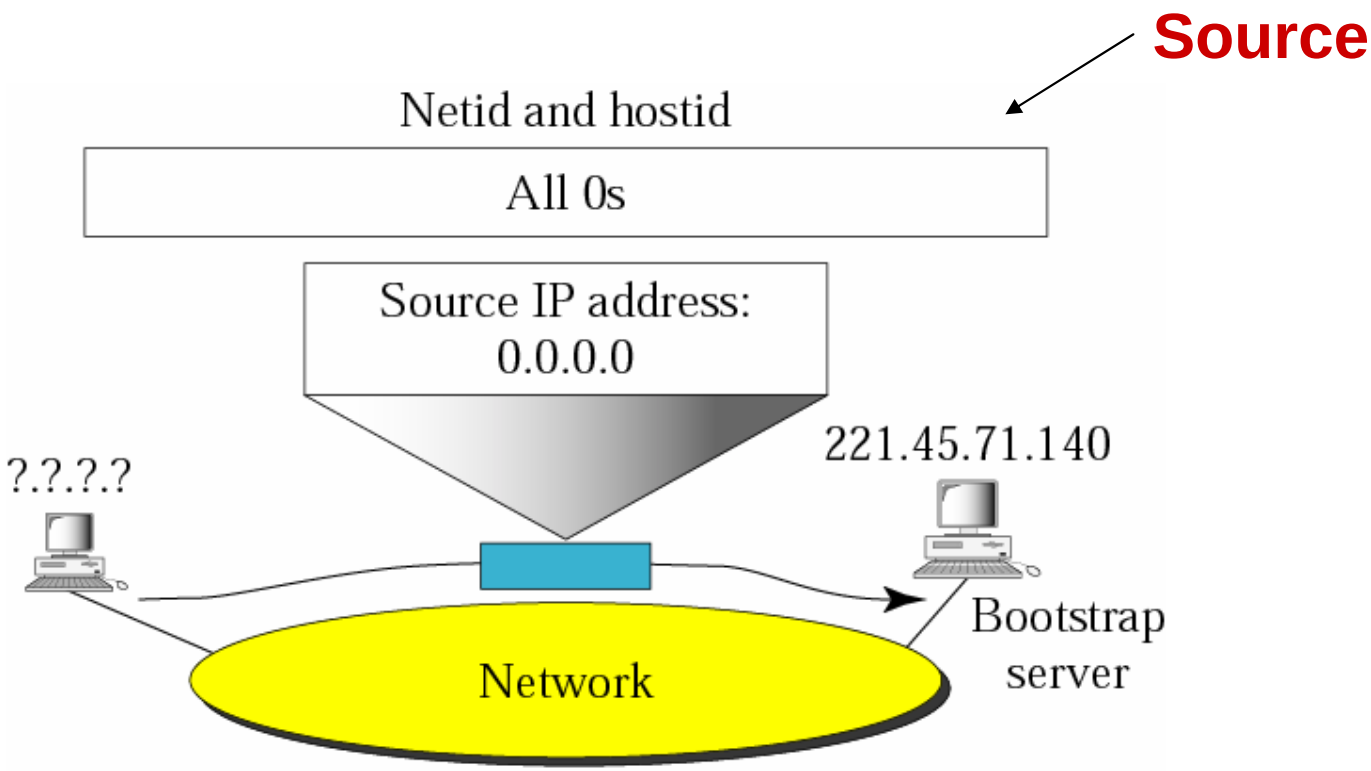
Example of *this* host on *this* address



A host that does not know its IP address uses the IP address 0.0.0.0 as the source address and 255.255.255.255 as the destination address to send a message to a bootstrap server.

Used by a host at boot time when it doesn't know its IP address

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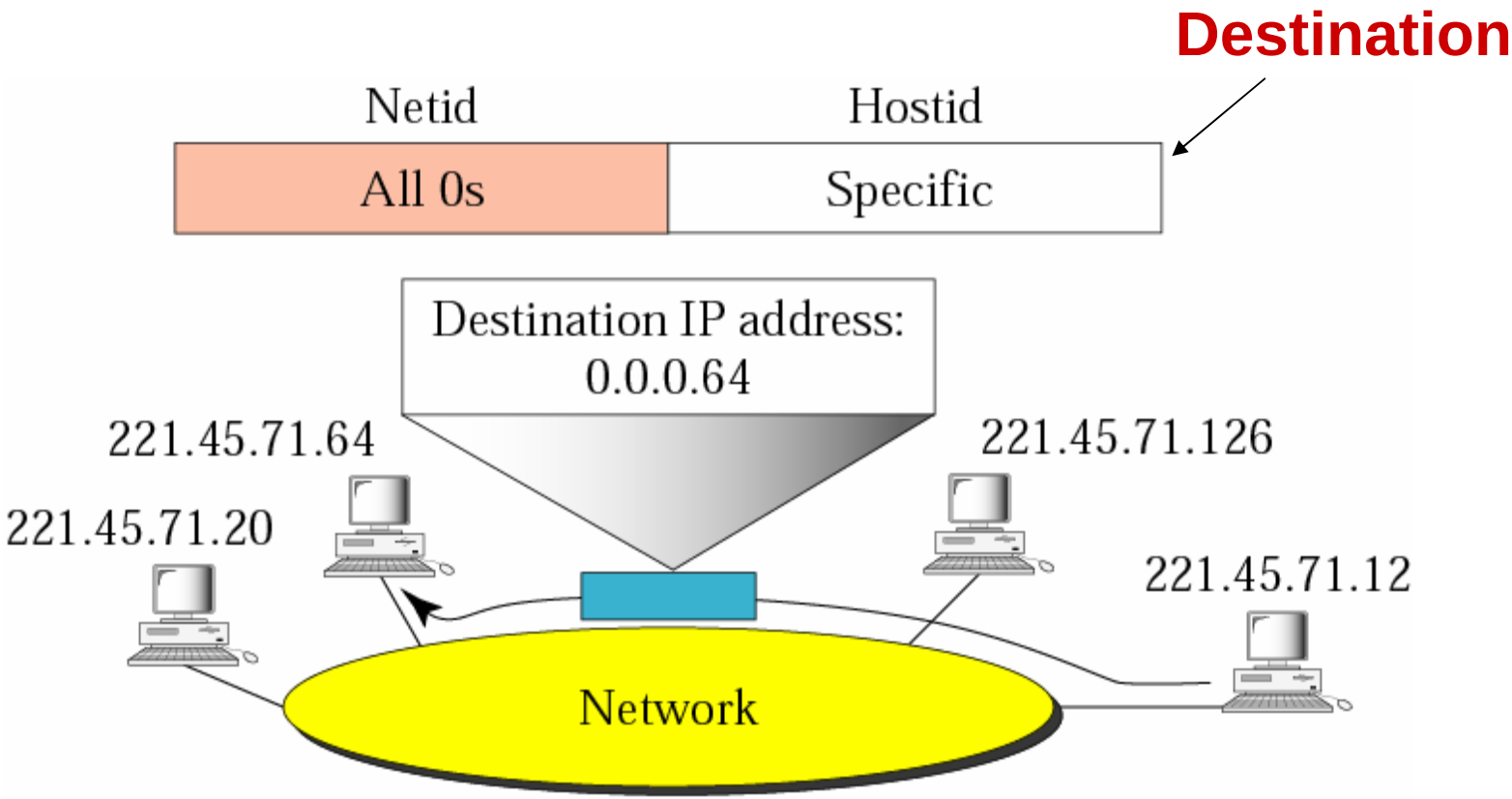
Example of specific host on *this* network



This address is used by a router or host
to send a message to a specific host on the same network.

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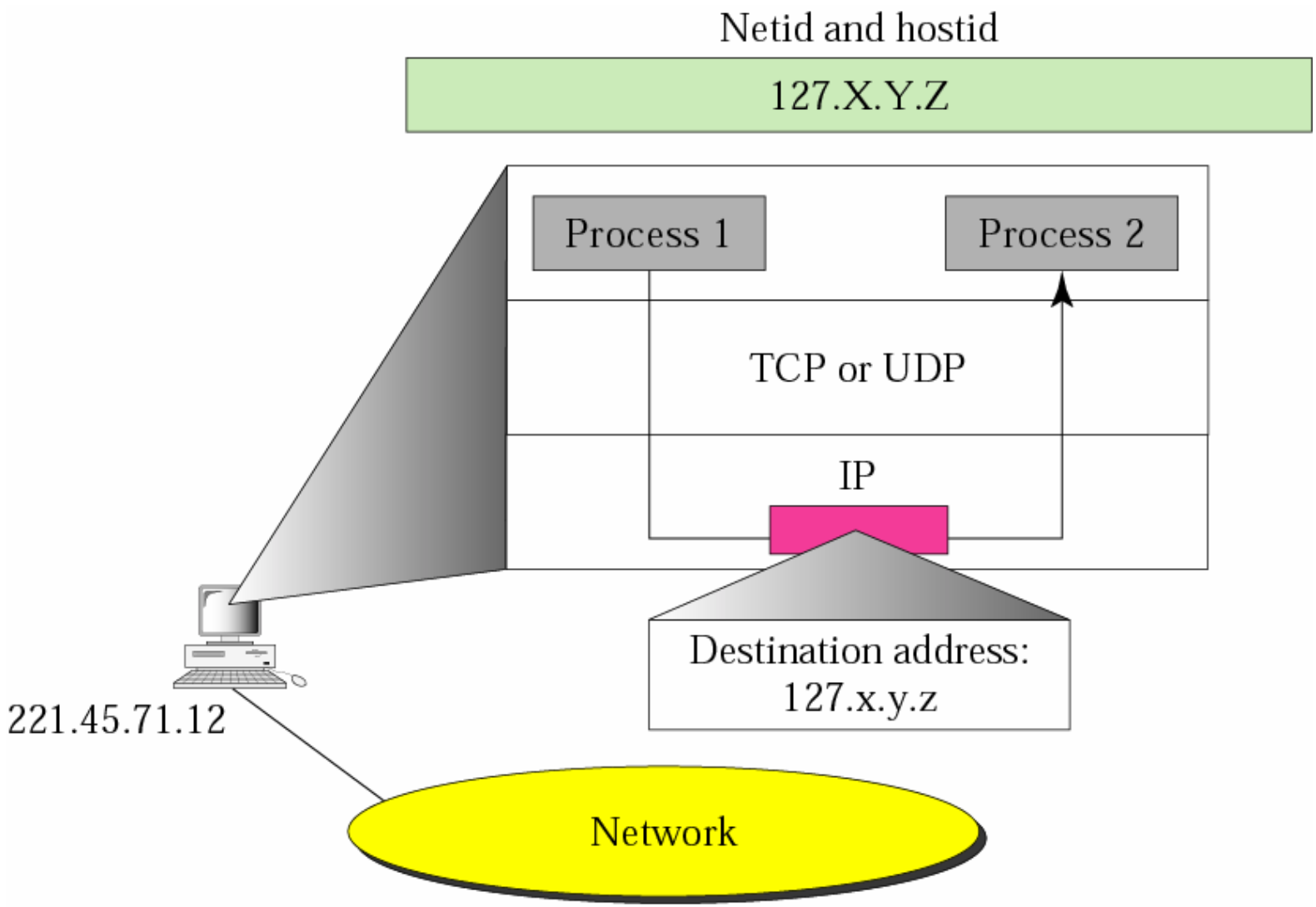
Example of loopback address



A packet with a loopback address will not reach the network.

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Unicast, Multicast, and Broadcast Addresses

Unicast communication is *one-to-one*.

Multicast communication is *one-to-many*.

Broadcast communication is *one-to-all*.

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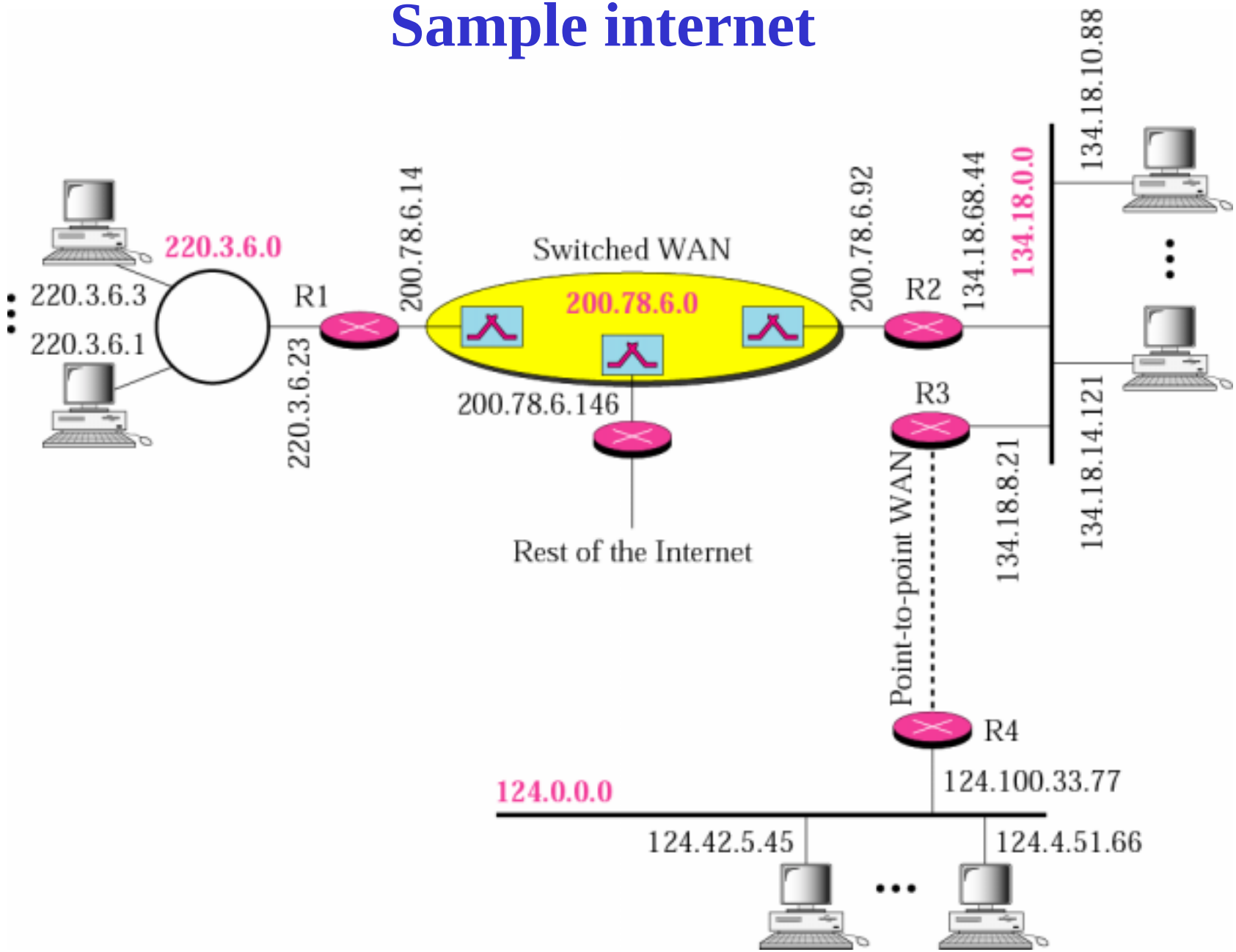
Sample internet



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